🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 · STANDARD 6.AT.9

Write and Represent Inequalities

Lesson 8-4 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 2 Standard

TODAY'S OBJECTIVES

Content Objective: I can write inequalities to represent real-world situations.

Language Objective: I can explain my inequality using the words inequality, greater than, less than, and at least / at most.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

Fill in each blank as we go. Use the Word Bank to help you.

WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Inequality | Greater than | Less than | At least / At most | Variable | No more than

Tap any word to see its meaning and a picture.

1. A math sentence comparing values with $<$, $>$, $\leq$, or $\geq$ is an ________.

2. The symbol $>$ that means one value is larger than another means ________.

3. The symbol $<$ that means one value is smaller than another means ________.

4. 'At least' means greater than or equal to, and 'at most' means ________ than or equal to.

5. A letter that stands for an unknown number is a ________.

6 The phrase that means less than or equal to is "no more than," written with the ______________ symbol.

Write About the Math

What speeds are possible for the getaway car? How did the phrase 'at most' help you write the inequality?

¿Qué velocidades son posibles para el carro de huida? ¿Cómo te ayudó la frase «como máximo» a escribir la desigualdad?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Inequality** — *Desigualdad*
- ☐ **Greater than** — *Mayor que*
- ☐ **Less than** — *Menor que*
- ☐ **At least / At most** — *Al menos / A lo más*
- ☐ **Variable** — *Variable*

Model: *The age could be 18 or any number greater, so it is not a single value.* — Students explain that 'at least 18' allows many values (18 and up), so an inequality with ≥ captures it while an equation would force one value.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **A speed of 65 mph is ____, a speed of 70 mph is ____, because 'at most 65' means the speed must be ____ 65.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Inequality
2. **Notes 2:** Greater than
3. **Notes 3:** Less than
4. **Notes 4:** At least / At most
5. **Notes 5:** Variable
6. **Notes 6:** No more than
7. **Watch for this mistake:** *A common mistake in Write Inequalities is using a strict symbol (> or <) for a phrase that actually includes the boundary value. For 'the suspect is at least 18 years old,' students often write $a > 18$ instead of the correct $a \geq 18$ — this wrongly excludes 18 itself, even though 'at least 18' means 18 counts as a match. The same error happens in reverse with 'at most': for 'the getaway bag holds at most 12 items,' students write $n < 12$ instead of $n \leq 12$. Always ask: does the key phrase include the number ('at least,' 'at most,' 'no more than,' 'no fewer than' → use ≥ or ≤) or exclude it ('more than,' 'fewer than' → use > or <)?*
8. **Try It 1:** A. $g \geq 12$ — *'At least 12' includes 12 and any number above it, so $g \geq 12$.*
9. **Try It 2:** A. $n < 5$ — *'Fewer than 5' does not include 5, so $n < 5$.*